Graduate Project Proposal: AI-Driven Intrusion Prevention System (IPS)

1. Project Title

Design and Implementation of an AI-Driven Intrusion Prevention System (IPS)

2. Introduction & Background

With the rise of cyberattacks, traditional firewalls and signature-based Intrusion Detection/Prevention Systems (IDS/IPS) are becoming insufficient to detect modern threats such as zero-day exploits, polymorphic malware, and insider attacks. This project aims to design and implement an **AI-powered IPS** that can **detect, prevent, and adapt to cyber threats in real time**, offering a more robust solution compared to conventional IPS technologies.

3. Problem Statement

Current IPS solutions are often:

- **Signature-dependent** → limited against unknown threats.
- **High in false positives** → leading to alert fatigue.
- **Difficult to scale** → with growing traffic and complex infrastructures.

Our IPS will address these issues by integrating **AI-driven anomaly detection, automated incident response, and adaptive prevention mechanisms**.

4. Project Objectives

1. **Develop a modular IPS architecture** with scalable deployment (cloud/on-premise).
2. **Integrate AI/ML algorithms** to detect anomalies and zero-day threats.
3. **Enable automated incident response** for containment and mitigation.
4. **Provide a web-based dashboard** for monitoring, reporting, and managing policies.
5. **Support DevOps and CI/CD pipelines** for continuous updates and scalability.

5. Scope of Work

Core Features

- **Traffic Monitoring & Analysis**: Capture and inspect packets in real time.
- **Signature-Based Detection**: Use open-source rule sets (e.g., Snort/Suricata).
- **AI-Powered Anomaly Detection**: ML/DL models for zero-day attacks.
- **Automated Prevention**: Blocking malicious traffic via firewall rules.
- **Incident Response Integration**: Alerting, logging, and automated containment.
- **Web Dashboard**: Visualization of attacks, system health, and reports.

6. Roles & Responsibilities

- **Cyber Security Team**
  - Penetration Testing: Simulate attacks to validate IPS effectiveness.
  - Incident Response: Define playbooks, response automation, and reporting.
- **AI Team**
  - Build ML/DL models for anomaly detection.
  - Train/test models on attack datasets (CIC-IDS, UNSW-NB15, etc.).
- **Web Developers**
  - Develop intuitive dashboard (attack visualization, logs, prevention rules).
  - Implement authentication and access control.
- **DevOps Team**
  - Containerization (Docker/Kubernetes).
  - CI/CD pipelines for continuous updates of signatures and ML models.
- **Networking Team**
  - Deploy IPS inline in a testbed environment.
  - Configure routing, VLANs, and traffic mirroring for testing.

7. Methodology

1. **Research & Requirement Gathering** – Analyze existing IPS systems (Snort, Suricata, Zeek).
2. **System Design** – Define architecture (network, AI, web, and DevOps layers).
3. **Dataset Collection & Preprocessing** – Use real traffic datasets (CIC-IDS2017, NSL-KDD).
4. **Model Development** – Train ML/DL models for anomaly detection.
5. **System Integration** – Combine detection engine, prevention module, and dashboard.
6. **Testing & Evaluation** – Simulate attacks (DoS, SQLi, brute force, etc.).
7. **Deployment** – Deploy on a hybrid cloud environment for scalability.

8. Tools & Technologies

- **Cybersecurity**: Snort, Suricata, Zeek, Metasploit, Wireshark
- **AI/ML**: Python, Scikit-learn, TensorFlow/PyTorch
- **Web Development**: React.js / Django / Node.js
- **DevOps**: Docker, Kubernetes, GitHub Actions, Jenkins
- **Networking**: Cisco Packet Tracer / GNS3 / pfSense

9. Expected Outcomes

- A working prototype of an **AI-driven IPS**.
- A **dashboard** showing real-time attack detection and prevention.
- Demonstration of **reduced false positives** compared to traditional IPS.
- A **documented framework** for deploying scalable IPS solutions.

11. Deliverables

1. Technical Report (Design + Implementation).
2. Working Prototype (IPS with AI + Dashboard).
3. Final Presentation & Demo.